Rabbit Alert Modeling Project

Sonit Ambashta

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1 Variables

- t: time (in steps, can adapt to days/minutes/seconds as needed)
- $A(t)$: number of alerted rabbits at time t
- k: alert propagation rate, measuring how quickly alerted rabbits cause additional rabbits to become alerted
- $\frac{dA}{dt}$: instantaneous rate of change of the alerted rabbit population.
- P: maximum number of alerted rabbits (capacity of the habitat)

2 Assumptions

- The rabbit population contains at least one rabbit.
- One or more rabbits may initially detect the predator.
- Alert propagation occurs through interactions between rabbits and is modeled as proportional to the number of rabbits that are already alerted.
- The alert propagation rate k is treated as a user-defined parameter because an empirically validated value is not currently available.
- All rabbits are equally capable of receiving and transmitting warning signals.
- Once alerted, a rabbit remains alerted for the duration of the simulation.
- Environmental effects such as terrain, vegetation density, weather, and visibility are ignored.
- Predator behavior and movement are not modeled.
- Time is treated as a continuous variable.
- The rabbit population is assumed to be well-mixed, meaning each alerted rabbit has an equal opportunity to influence any other rabbit.
- The model is intended for wild rabbit populations and does not account for behaviors associated with domesticated rabbits.
- Differences in alert behavior among rabbit species are not modeled. All wild rabbits are treated as having identical alert-propagation characteristics.

3 Differential Equation and Approach

$$\frac{dA}{dt} = kA$$

Note: this is a separable differential equation

- multiply dt on both sides

$$dA = kAdt$$

- divide A on both sides

$$\frac{1}{A}dA = kdt$$

- integrate both sides (don't forget the constant of integration)

$$\int \frac{1}{A} dA = \int k dt$$

$$ln|A| = kt + C$$

- isolate A

$$A = e^{kt+C}$$

- use exponent rules to separate the C

$$A = e^C e^{kt}$$

- let e^C be an arbitrary constant C_1

$$A = C_1 e^{kt}$$

- apply the initial condition $A(0) = A_0$

$$A_0 = C_1 e^{k(0)}$$

- simplify since $e^0 = 1$

$$A_0 = C_1$$

- substitute $C_1 = A_0$ back into the solution

$$A = A_0 e^{kt}$$

3.1 Example Model

Suppose a user provides

$$k = 0.7$$

and

$$A_0 = 2.$$

The differential equation becomes

$$\frac{dA}{dt} = 0.7A.$$

Using the solution derived previously,

$$A(t) = A_0 e^{kt},$$

we substitute the values of A_0 and k:

$$A(t) = 2e^{0.7t}.$$

This function models the number of alerted rabbits over time assuming that each alerted rabbit contributes proportionally to the spread of awareness throughout the population. The corresponding Python code is:

```
>>> equation = form_differential_equation(0.7)
>>> solution = solve_equation(equation, 2)
>>> print(solution)
```

Output:

```
Eq(A(t), 2*exp(0.7*t))
```

Some example evaluations are:

$$A(0) = 2e^0 = 2$$
$$A(1) = 2e^{0.7} \approx 4.03$$
$$A(2) = 2e^{1.4} \approx 8.11$$
$$A(3) = 2e^{2.1} \approx 16.33$$

These values illustrate the exponential nature of the model, where the number of alerted rabbits grows increasingly rapidly over time. This behavior motivates the introduction of a population cap in later versions of the model.

3.2 Notes

- This equation is for any arbitrary k that will be plugged in (user adjustable), however growth cannot be demonstrated for $k = 0$ since that would signify no change.
- Right now, there is not a supporting rate at which rabbits get alerted, so this serves as a user-adjustable cli interface that allows adding a rate
- A certain number of rabbits can be alerted at a moment of time (especially when they are together), so the initial value setup can also be provided by the user
- In this model, it is also assumed that there is no upper bound on the number of rabbits that can be alerted at any given time.

4 Major Issue with this Approach and Proposed Solution

The original model assumed an unlimited rabbit population, producing unbounded exponential growth. This assumption was later relaxed through the introduction of a carrying-capacity parameter P, resulting in a capped exponential model. The idea is to allow exponential growth until the number of alerted rabbits reaches the maximum number of rabbits in the habitat. To determine when this occurs, solve for the point of intersection between the exponential solution and the carrying capacity.

$$A(t) = A_0 e^{kt}$$

$$P = A_0 e^{kt}$$

Divide both sides by A_0:

$$\frac{P}{A_0} = e^{kt}$$

Take the natural logarithm of both sides:

$$\ln\left(\frac{P}{A_0}\right) = kt$$

Solve for t:

$$t^* = \frac{\ln(P/A_0)}{k}$$

where t^* represents the time at which the alerted population reaches the habitat capacity. Once this point is reached, there are no additional rabbits available to become alerted. Therefore, the rate of change becomes zero. This gives the piecewise differential equation

$$\frac{dA}{dt} = \begin{cases} kA, & A < P, \\ 0, & A \geq P. \end{cases}$$

The corresponding piecewise solution is

$$A(t) = \begin{cases} A_0 e^{kt}, & t < t^*, \\ P, & t \geq t^*. \end{cases}$$

where

$$t^* = \frac{\ln(P/A_0)}{k}.$$

5 Inspiration and Rationale

- This project began as an opportunity to apply differential equations outside of a classroom setting and explore how mathematical models can be used to represent real-world phenomena programmatically. Rabbits were chosen as the motivating domain because they are an important component of many ecosystems and serve as prey for a wide range of predators. Their survival often depends on the rapid detection and communication of threats, making predator-awareness propagation a natural phenomenon to explore through differential equations. The initial goal was not to produce a biologically validated model. Rather, the objective was to practice the process of mathematical modeling: defining assumptions, constructing a differential equation, solving the equation analytically, implementing the solution in Python, and evaluating the realism and limitations of the resulting model.
- As development progressed, the project evolved beyond simply solving an exponential differential equation. The model was extended to include a finite habitat capacity after it became clear that unrestricted exponential growth implied an unrealistic infinite rabbit population. This led to the introduction of a capped exponential model and further discussion regarding alternative modeling approaches and their underlying assumptions.

- Although a validated behavioral propagation rate is not currently available, the framework is designed as a command-line utility for exploring different parameter combinations. The project may serve as an educational tool for learning differential equations and mathematical modeling, while also providing a foundation for future investigations into rabbit behavior, ecosystem dynamics, and animal communication.